

Ahmed Amr

Machine Learning & Computer Vision Engineer

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[LinkedIn](#) • [GitHub](#) • [Portfolio](#)

Profile

I help businesses unlock the true potential of their data by delivering end-to-end intelligent solutions. Specializing in deep Exploratory Data Analysis (EDA), dynamic Data Visualization, and robust Machine Learning & Deep Learning architectures, I turn complex raw data into actionable, scalable AI products. Fresh Computer Science graduate skilled in Python, PyTorch, TensorFlow, and OpenCV.

Education

Bachelor of Computer Science at Modern Academy Maadi, Cairo, Egypt

September 2022 - Present

Internships

Data Science & Machine Learning Intern at [IT-Gate Academy](#), Cairo, Egypt

July 2024 - August 2024

Completed intensive training in Data Science fundamentals and practical machine learning. Gained hands-on experience in exploratory data analysis, data preprocessing, feature engineering, and building & evaluating ML models on real-world dataset

Skills

Programming & Databases: Python, SQL

Data Science & Analysis: NumPy, Pandas, SciPy, Scikit-learn, Matplotlib

AI & Computer Vision: PyTorch, OpenCV, YOLO, U-Net, Image Segmentation, Object Detection

Backend & Tools: Flask, Docker, Git, GitHub

Projects

Axon: End-to-End Alzheimer's Detection System • [GitHub](#) • [Live Demo](#)

January 2026 - June 2026

Designed and developed an end-to-end medical AI solution classifying brain MRI scans into 4 stages of Alzheimer's Disease with 93.2% accuracy. Engineered a custom CNN architecture using TensorFlow/Keras, trained on 33,984 augmented MRI images with BatchNorm, Dropout, and MobileNetV2 normalisation. Achieved 99% precision/recall on Moderate Demented cases, critical for urgent medical intervention. Built a Flask REST API for real-time inference and integrated it with a Flutter mobile application. Deployed the model to Hugging Face Spaces using Gradio for public accessibility

Customer Churn Prediction • [GitHub](#)

May 2026

Built 6 ML models (both built-in and from scratch) to predict customer churn on 440K records. Conducted deep EDA and feature engineering. Best model: Random Forest at 93% accuracy.

Food Image Classifier • [GitHub](#) • [Live Demo](#)

May 2026

Fine-tuned ResNet50 on the Food-101 dataset (101K images, 101 classes) for food classification and calorie estimation. Deployed as an interactive web app on Hugging Face Spaces.

Certificates

Statistics 101 from Cognitive Class • [Link](#)

November 2025

Data Analysis with Excel Pivot Tables from 365 Data Science • [Link](#)

November 2025

Scientific Computing with Python from freeCodeCamp • [Link](#)

March 2026